THE DERIVED PULLBACK IS COMPATIBLE WITH COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the reduction and the comparison are both inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Derived pullback along a morphism of ringed topoi. Same chapter, the construction of Lf$\widehat{\ }$* and its identification with the naive pullback on K-flat complexes with flat terms. Supplies the functors composed in the statement and the representatives with which the comparison is made.
- Composition of morphisms of ringed topoi and the naive pullback functors. Same chapter, the identity f$\widehat{\ }$*g$\widehat{\ }$* = (g f)$\widehat{\ }$* for the naive pullbacks, together with the stability of K-flatness with flat terms under pullback. Supplies the comparison at the level of representatives.

Source reference. Original lemma 06YS: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3207–3207. Navigation pointer invoked at proof line 3616; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06YY: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3585–3585. Navigation pointer invoked at proof line 3674; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 0G7E: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3508–3508. Navigation pointer invoked at proof line 3632; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Consider morphisms of ringed topoi $f : (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C}) \to (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D})$ and $g : (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D}) \to (Sh(\mathcal{E}), \mathcal{O}_\mathcal{E})$. Then $Lf^* \circ Lg^* = L(g \circ f)^*$ as functors $D(\mathcal{O}_\mathcal{E}) \to D(\mathcal{O}_\mathcal{C})$.*

Proof. Let E be an object of $D(\mathcal{O}_\mathcal{E})$. We may represent E by a K-flat complex $\mathcal{K}^\bullet$ with flat terms, see Lemma 06YS. By construction Lg^*E is computed by $g^*\mathcal{K}^\bullet$, see Lemma 06YY. By Lemma 0G7E the complex $g^*\mathcal{K}^\bullet$ is K-flat with flat terms. Hence Lf^*Lg^*E is represented by $f^*g^*\mathcal{K}^\bullet$. Since also $L(g \circ f)^*E$ is represented by $(g \circ f)^*\mathcal{K}^\bullet = f^*g^*\mathcal{K}^\bullet$ we conclude. □